

Thermal expansion

$$\Delta l = \alpha l_0 \Delta T$$

$$\Delta A = 2\alpha A_0 \Delta T$$

$$\Delta V = \beta V_0 \Delta T = 3\alpha V_0 \Delta T$$

Calorimetry

$$Q = mc\Delta T = nC\Delta T \text{ (no phase transition)}$$

$$Q = mL \text{ (for a phase transition)}$$

$$\sum_{i=1}^n Q_i = 0 \text{ (for a closed system)}$$

Heat conduction

$$H = \frac{dQ}{dt} = -kA \frac{dT}{dx}$$

Ideal gases

$$PV = Nk_B T = nRT$$

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

$$f_{\text{Maxwell}}(v) = 4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

$$\Delta E_{\text{int}} = Q - W$$

$$dE_{\text{int}} = dQ - PdV$$

$$W = \int PdV$$

$$E_{\text{int}} = \frac{d}{2} Nk_B T = nC_V T$$

$$C_V = \frac{d}{2} R$$

$$C_P - C_V = R = N_A k_B$$

$$\gamma = \frac{C_P}{C_V} = \frac{d+2}{d}$$

$$e = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$$

$$e_{\text{ideal}} = 1 - \frac{T_L}{T_H}$$

$$S = \int \frac{dQ}{T} \text{ (For reversible processes)}$$

$$dQ = TdS$$

$$\Delta S_{\text{syst}} + \Delta S_{\text{env}} > 0 \text{ (For irreversible processes)}$$

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\vec{F} = Q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\lambda = \frac{dQ}{dl} ; \quad \sigma = \frac{dQ}{dA} ; \quad \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\Delta U = q\Delta V$$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{\text{eq}} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$\epsilon = K\epsilon_0$$

$$C_D = KC_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2} |\vec{E}|^2 \text{ (energy density)}$$

	Q	W
Isobaric ($P=const$)	$nC_p\Delta T$	$P\Delta V$
Isochoric ($V=const$)	$nC_v\Delta T$	0
Isothermal ($PV=const$)	$nRT \ln\left(\frac{V_f}{V_0}\right)$	$nRT \ln\left(\frac{V_f}{V_0}\right)$
Adiabatic ($PV^\gamma = const$)	0	$-\frac{d}{2}(P_f V_f - P_0 V_0)$

C_v, C_p and $\gamma = C_p/C_v$ for an Ideal Gas

1. For a monoatomic gas, $d=3$

$$C_v = \frac{3R}{2}, C_p = \frac{5R}{2} \text{ and } \gamma = \frac{5}{3} = 1.67$$

2. For a diatomic gas, $d=5$

$$C_v = \frac{5R}{2}, C_p = \frac{7R}{2} \text{ and } \gamma = \frac{7}{5} = 1.4$$

3. For a triatomic or polyatomic gas, $d=6$

$$C_v = 3R, C_p = 4R \text{ and } \gamma = \frac{4}{3} = 1.33$$

Current

$$I = \frac{dQ}{dt}$$

$$I = \int \vec{j} \cdot d\vec{A}$$

$$\vec{j} = nQ\vec{v}_d = \frac{\vec{E}}{\rho}$$

Resistance

$$R = \int \rho \frac{dl}{A}$$

$$\rho(T) = \rho(T_0)(1 + \alpha(T - T_0))$$

$$R_{eq} = R_1 + R_2 \text{ (In series)}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \text{ (In parallel)}$$

DC circuits

$$P = IV$$

$$V = IR \text{ (Ohm's law)}$$

$$\sum_{junc.} I_{in} = \sum_{junc.} I_{out} \text{ (junction rule)}$$

$$\sum_{loop} V_i = 0 \text{ (loop rule)}$$

$$\text{Charging capacitor: } \frac{dQ}{dt} + \frac{Q}{RC} = \frac{\mathcal{E}}{R}$$

$$Q(t) = C\mathcal{E}(1 - e^{-t/(RC)}) \text{ and } I(t) = \frac{dQ}{dt}$$

with $C\mathcal{E} = Q_{max}$ and $\tau = RC$

$$\text{Discharging capacitor: } \frac{dQ}{dt} + \frac{Q}{RC} = 0$$

$$Q(t) = C\mathcal{E}e^{-t/(RC)} \text{ and } I(t) = -\frac{dQ}{dt}$$

Magnetism

$$d\vec{F}_m = I d\vec{l} \times \vec{B}$$

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{\mu} = NI\vec{A}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Integrals

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^\infty x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int (1+x^2)^{-1} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \ln(\sqrt{1+x^2} + x)$$

$$\int x(1+x^2)^{-\frac{1}{2}} dx = \sqrt{1+x^2}$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{x} dx = \ln(x)$$

$$\int \frac{1}{\cos(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right|\right)$$

$$\int \frac{1}{\sin(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2}\right)\right|\right)$$

Electromagnetic Induction

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$M = N_1 \frac{\Phi_1}{I_2} = N_2 \frac{\Phi_2}{I_1}$$

$$L = N \frac{\Phi_B}{I}$$

$$U = \frac{1}{2} LI^2$$

$$U = \int \frac{1}{2\mu_0} |\vec{B}|^2 dV$$

Gradient and displacement

- *Cylindrical coordinates :*

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z}$$

- *Spherical coordinates :*

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

Lengths, Areas and Volumes

$$\text{Arc length: } dL = R d\theta$$

$$\text{Cylinder: } A = 2\pi R H ; V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Thin Circular Ribbon : } dA = 2\pi R dR$$

$$\text{Section of Thin Ribbon : } dA = R dR d\theta$$

$$\text{Sphere: } A = 4\pi R^2 ; V = \frac{4}{3} \pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$

Taylor Expansions near x=0

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2}mv^2$$

$$\vec{F}_{sp} = -k \vec{x} ; U_{sp} = \frac{1}{2}kx^2$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2\cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$+ \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \text{ (law of cosine)}$$